

H4 — ML + Embedded Track

Member: _____

Your role

You own the runtime memory-layout map and the shipping-format (group 128) quantization. You write the model card.

Weekly tasks

Week 1 — Memory layout map

- Task: Read the src/ runtime. Map the memory tiers: SRAM, PSRAM, flash. Locate the KV cache, the int8 head, and the PLE table. Compare with the published numbers.
- Deliverable: The 1-page memory-layout map.
- Verify: You can explain why the head sits in PSRAM and the table in flash.
- Learn: Neural network basics. Watch the 3Blue1Brown series on neural networks.

Week 2 — Quantize harness

- Task: Build the quantize harness notebook for the shipping format: group 128, fp16 scales.
- Deliverable: The quantize harness notebook in the shared Drive.
- Verify: The notebook runs `quantize_eval` on one seed.
- Learn: Embeddings and next-token prediction. Explain the 4-bit weight format.

Week 3 — Group-128 PTQ

- Task: Run the shipping-format quantize eval for both seeds: `uv run python -m research.tinystories.quantize_eval --tag cleandeploy --seed 0 --group 128 --fp16-scales` and `--seed 1`.
- Deliverable: The shipping-format degradation table.
- Verify: The table matches the published shape: baseline degrades more than ple.
- Learn: The transformer block. Explain why group size matters.

Week 4 — Model card

- Task: Write the host-side model card: our trained models, sizes, ppl, sample outputs.
- Deliverable: MODEL_CARD.md in the club repo.
- Verify: Every number in the card appears in the tracking sheet.
- Learn: Perplexity and evaluation. Explain what a model card documents.

Week 5 — Use-case shortlist

- Task: Write the use-case shortlist: 3-5 practical candidates with feasibility notes.
- Deliverable: USE_CASES.md in the club repo.
- Verify: Each candidate lists the model limit it respects.
- Learn: RoPE and KV cache. Explain why context length matters for a use case.

Week 6 — Use-case decision

- Task: Write the use-case decision doc: one recommended use case with justification.
- Deliverable: USE_CASE_DECISION.md in the club repo.
- Verify: H5 presents the decision to the team and collects feedback.
- Learn: Fine-tuning and PTQ. Explain what the use case needs from the model.

Week 7 — Teach perplexity

- Task: Present perplexity and evaluation to the team at the meetup.
- Deliverable: Your teach-back slides.
- Verify: R1 can compute ppl from a loss value.
- Learn: Prepare your presentation deeply.

Week 8 — Retrospective summary

- Task: Write the retrospective summary: what worked, what did not.
- Deliverable: RETROSPECTIVE.md in the club repo.
- Verify: Every member adds one line to the retro.
- Learn: Write the "what I learned" page.

Learning path

1. Neural network basics: perceptron, MLP, gradient descent.
2. Embeddings, tokenization, next-token prediction.
3. Transformer block: attention, LayerNorm, residuals, MLP.
4. Loss, perplexity, sampling, generation.
5. Attention math, RoPE, KV cache. PLE paper.
6. Fine-tuning, transfer learning, PTQ.
7. Teach one topic to the team.
8. Write the "what I learned" page.

Resources

- Repo: <https://github.com/slvDev/esp32-ai>
- src/: <https://github.com/slvDev/esp32-ai/tree/main/src>
- 3Blue1Brown neural networks: https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi